

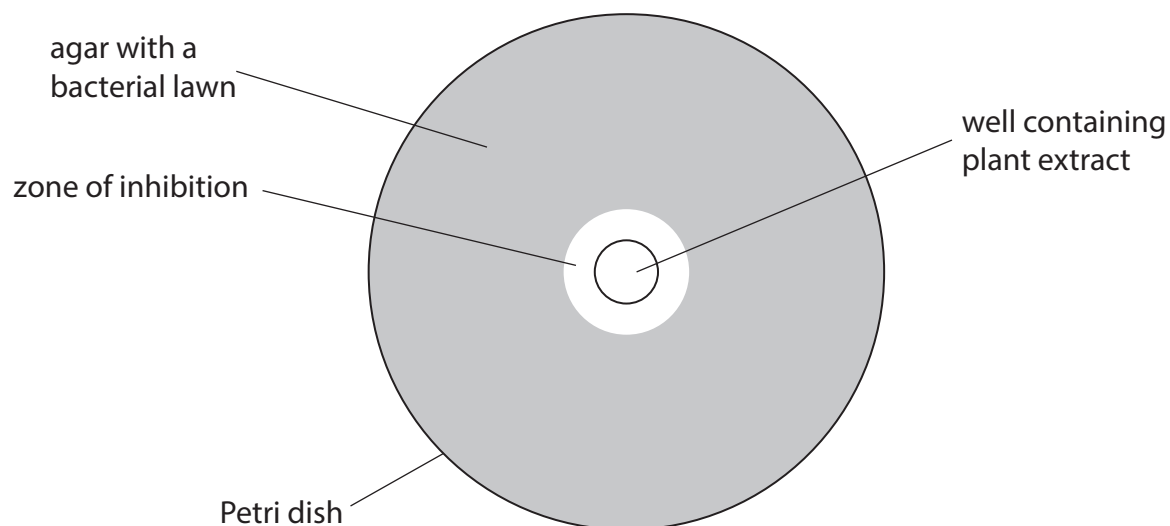
- 3 Petri dishes with nutrient agar were used in an investigation of the antimicrobial properties of plant extracts.

Bacteria were spread on the surface of the agar to form a lawn.

Plant extracts were placed in wells cut in the agar.

After incubation, the size of the zone of inhibition is a measure of the antimicrobial effect of the plant extract.

The diagram shows a Petri dish after the incubation period.



- (a) (i) Explain why the dishes were incubated for 48 hours at 30°C.

(2)

48 hours

30°C

- (ii) Explain **three** aseptic techniques that should be used when transferring bacteria from one culture to another.

(3)

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- (b) The technique described was used to investigate the antimicrobial effects of oil A and oil B, extracted from two species of plant.

Different bacterial lawns were made using *E. coli* and *B. subtilis*.

- (i) Explain how the area of the zone of inhibition could be measured accurately.

(2)

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(ii) State **one** independent variable in this investigation.

(1)

(iii) The table shows the results from this investigation.

Type of bacteria	Diameter of zone of inhibition / mm	
	oil A	oil B
<i>B. subtilis</i>	20.66	11.33
<i>E. coli</i>	8.66	0.00

Determine the effect of these two oils on the growth of these bacteria.

(4)



(c) The size of the zones of inhibition may also be affected by the rate of diffusion of the antimicrobial molecules through the agar.

- (i) Suggest **two** factors that may affect the rate of diffusion of the antimicrobial molecules.

(2)

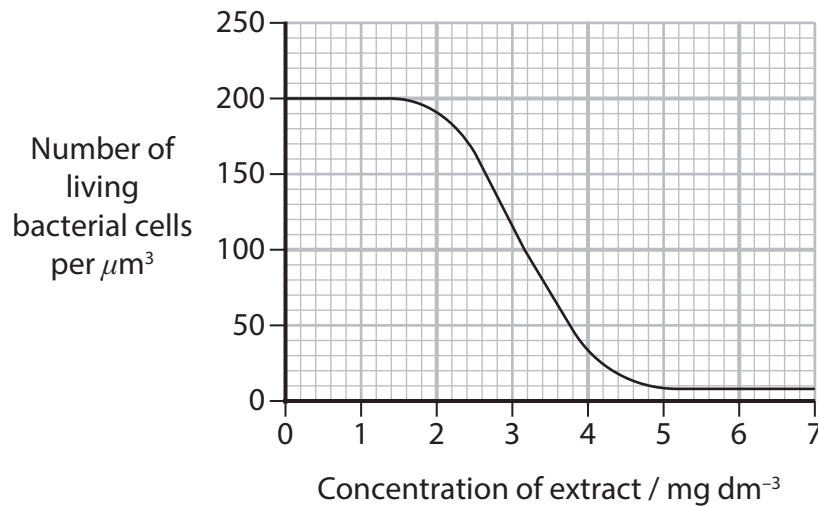
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- (ii) A method that avoids this problem is to find the minimum inhibitory concentration (MIC) of the extract.

The MIC can be defined as the minimum concentration of the antimicrobial molecule that will reduce the number of living bacterial cells by 90%.

In an investigation the following data were obtained.



Determine the MIC for this extract.

(2)

Answer mg dm^{-3}

(Total for Question 3 = 16 marks)

TOTAL FOR PAPER = 50 MARKS

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